

Exp No: 13 EMBEDDED C PROGRAM TO ADD AN ARRAY OF 16-BIT NUMBERS AND STORE THE 32-BIT RESULT IN INTERNAL

DATE: 13/08/2026

C:\Keil_v5\C51\Examples\HELLO\EMBEDDED C\EXP11\EXP13.uvproj - µVision

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Project: EXP13

Target 1

Source Group 1

exp13.c

```

1 #include<reg51.h>
2 void main(void)
3 {
4     unsigned int i;
5     unsigned int array[5]={0x1111,0x2222,0x8888,0x4444,0xABCD};
6     unsigned long sum=0;
7     for(i=0;i<5;i++)
8     {
9         sum=sum+array[i];
10    }
11    P0=(unsigned char) (sum&0xFF);
12    P1=(unsigned char) ((sum>>8)&0xFF);
13    P2=(unsigned char) ((sum>>16)&0xFF);
14    while(1);
15 }

```

Build Output

linking...

Program Size: data=23.0 xdata=0 code=404

".\Objects\EXP13" - 0 Error(s), 0 Warning(s).

Build Time Elapsed: 00:00:00

OUTPUT:

C:\Keil_v5\C51\Examples\HELLO\EMBEDDED C\EXP11\EXP13.uvproj - µVision

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Registers

Register Value

Register	Value
R0	0x00
R1	0x00
R2	0x00
R3	0x00
R4	0x00
R5	0x00
R6	0x00
R7	0x00
SP	0x00
DPTR	0x0000
PC	0x0009
STATUS	0x0000
SEC	0x000194...
PSW	0x00

exp13.c

```

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```

Parallel Port 0

Port 0

P0: 0xCC 7 Bits 0

Parallel Port 1

Port 1

P1: 0xAB 7 Bits 0

Parallel Port 2

Port 2

P2: 0x01 7 Bits 0

Command

Running with Code Size Limit: 2K

Load "C:\Keil_v5\C51\Examples\HELLO\EMBEDDED C\EXP11\Objects\EXP13"

Watch 1

Name	Value	Type
sum	0x001ABCC	ulong
array	D:0x08	array[5] of uint
0x00	0x0000	int
Address: 0x00	<cannot evaluate>	uchar
Address: 0x00	<cannot evaluate>	uchar

ASM ASSIGN BreakDisable BreakEnable BreakKill BreakList BreakSet BreakAccess COVTOFILE

Simulation

t1: 280.97110800 sec L5 C:1 CAP: NUM 51